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A 64-Gb/s 0.33-pJ/bit PAM4 Receiver Analog Front-End with a Single-Stage Triple-Peaking CTLE Achieving 22.5- dB Boost in 40-nm CMOS Process

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ABSTRACT This work presents a PAM4 receiver analog frontend (AFE) operating up to 64 Gb/s. The electronic integrated circuit (EIC) is fabricated in 40-nm CMOS technology. This AFE is composed of a single-stage Continuous-Time Linear Equalizer (CTLE), a Variable Gain Amplifier (VGA), an input impedance matching network, a buffer stage, and an output buffer. The single-stage triple-peaking CTLE proposed employs current reuse technique and a multi-feedback structure, enabling the adjustment of peaking in the low, mid, and high-frequency bands. Thus, only one-stage CTLE is sufficient to achieve an over-20-dB boost at Nyquist frequency to save power. The VGA adopts an enhanced structure based on the Gilbert cell, where the gain is manipulated by controlling the gate voltage of MOS transistors. The CTLE undergoes variations in its DC gain during the adjustment process to equalize channel losses. The role of the VGA is to stable the DC gain changes induced by the adjustment of the CTLE. The output buffer adopts two stages, aiming to ensure that the gain does not attenuate excessively while maintaining output impedance matching. The AFE consumes 21.1mW with a supply voltage of 1.5/1V. It can provide a maximum boost of 22.5 dB, and the data rate reaches up to 64 Gb/s. Additionally, it features peaking adjustment capabilities in the low, mid, and high-frequency bands. Finally, the measurement demonstrates its ability to effectively equalize a channel with a 12-dB loss at the Nyquist frequency of 16 GHz.

INDEX TERMS Analog front-end (AFE), continuous-time linear equalizer (CTLE), VGA, Triple Peaking, PAM4, low-power

I. INTRODUCTION

In recent years, with various breakthrough applications, such as cloud computing, the Internet of Things (IoT), artificial intelligence, etc., the data throughput of data centers has been steadily increasing year by year[1,2]. Pulse-amplitude modulation (PAM-4) signaling continues to be the preferred modulation scheme, primarily due to its superior spectral efficiency compared to non-return-to-zero (NRZ) modulation and its more straightforward implementation compared to more complex schemes[3]. Therefore, enhancing the energy efficiency of the widely utilized PAM4 serial link transceiver, crucial for data transmission within data centers, holds significant importance in addressing the power consumption challenges posed by the expanding scale of data centers[4,5]. In PAM4 serial link transceivers, a substantial portion of power consumption is attributed to the analog front-end (AFE) circuit. This is due to its requirement to compensate for channel loss through the continuous-time linear equalizer (CTLE) and amplify the swing of the CTLE output to several hundreds of mV. Therefore, the low-power AFE circuit holds significant importance in the design of low-power PAM4 serial link transceivers.

The primary function of the AFE is to equalize the losses of the channel. We understand that channels introduce high-frequency losses due to phenomena like skin effect, which are non-ideal characteristics. However, for channels in certain mediums, significant attenuation can also occur at low and mid frequencies. This necessitates the CTLE in the AFE circuit to have adjustable compensation capabilities across low, mid, and high-frequency bands for better equalization of channel losses. In previous works, conventional CTLE only achieves high-frequency peaking by reducing DC gain. In order to increase the equalization capability of CTLE, low-frequency feedback technology is used to weaken the DC gain before fixing low-frequency, and then high-frequency feedforward technology is used to achieve high-frequency boost adjustment. Combining technologies above can achieve peaking adjustment at low, medium and high frequency. However, the topology need for multi-stage CTLE cascading, leading to higher power consumption[6,7].

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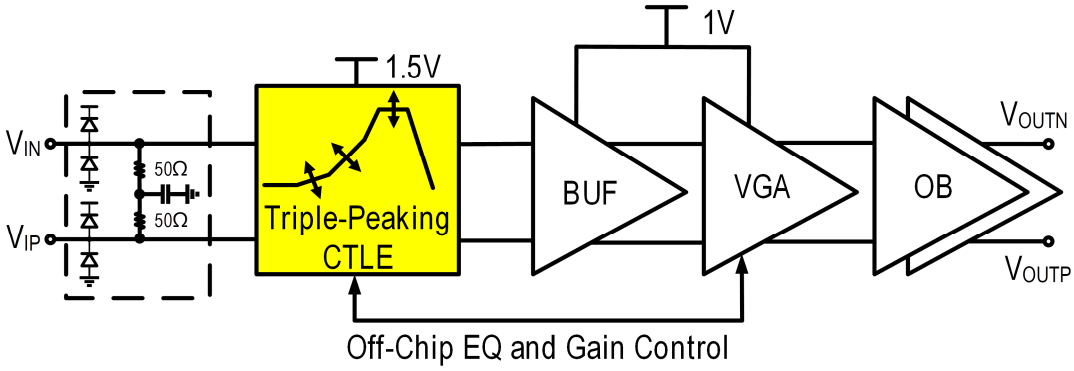


FIGURE 1. **The Proposed Receiver AFE Architecture.**

To surmount this issue, in this article, we present a 64-Gb/s PAM4 receiver AFE designed in a 40-nm CMOS process, achieving extremely low energy efficiency of 0.33pJ/bit[8]. To reduce AFE power consumption, which was predominantly associated with multi-stage CTLE in prior research, we present a groundbreaking solution—a single-stage triple-peaking CTLE (STP-CTLE). This inventive design integrates a current reuse technique, enabling the manipulation of three peaks (low, middle, and high-frequency peaking) with only a single stage. A buffer is connected after the CTLE to stabilize the output common-mode voltage, and simultaneously, it can drive larger input capacitance in the subsequent VGA. The VGA is an improved version based on the Gillbert cell structure. In contrast to the traditional Gilbert cell, which adjusts gain by varying the current source magnitude, the proposed VGA modifies gain by altering the gate voltage of the MOS transistors in series with the load resistance. This structure exhibits superior linearity, making it particularly advantageous for PAM4 signals that demand high linearity[9]. The output buffer employs a two-stage buffer structure, enabling impedance matching without compromising the overall gain performance of the AFE.

II. Receiver AFE Architecture

Fig. 1 shows the overall block diagram of the AFE, consisted of an input impedance matching network, a STP-CTLE for compensating channel losses, a VGA for gain control, a buffer isolating between CTLE and VGA, and an output buffer for testing purposes.

The STP-CTLE can achieve a maximum channel compensation of 22.5dB. It has adjustable compensation capabilities across low, mid, and high-frequency bands. Additionally, due to the adoption of current reuse technology, its power consumption is only 12.6mW. The gain adjustment range of the VGA is 5.8dB. As the CTLE adjusts its parameters to compensate for channel variations, there are changes in the DC gain of the CTLE. To counteract this, we appropriately adjust the parameters of the VGA to ensure that the overall gain, considering both the VGA and CTLE, remains constant. The addition of a buffer between VGA and CTLE serves two purposes. Firstly, to accommodate the need for VGA to provide a larger gain, its input transistors have larger dimensions, resulting in increased parasitic capacitance at the input. Introducing a buffer enhances the drive capability of the preceding stage to VGA, ensuring sufficient bandwidth. Secondly, when CTLE adjusts the magnitude of high-frequency peaking, it alters the DC output operating point. The inclusion of a buffer is intended to stabilize the DC operating point. Additionally, the output buffer (OB) is used for testing, which employs a highfrequency peaking to compensate for the loss of testing cable.

III. Proposed Single-Stage Triple-Peaking CTLE

In previous Analog Front-End (AFE) designs, achieving three peaks necessitated the use of at least two cascaded CTLEs, as depicted in Fig 2. The first stage could employ a high-frequency feedforward CTLE (HFF-CTLE), integrating a degenerated RC of the source and a high-frequency feedforward topology to achieve mid and high-frequency peaking. Meanwhile, the second stage, utilizing a low-frequency equalization (LFEQ) [10], could incorporate a feedback loop with a low-pass filter for low-frequency peaking. As illustrated in Figure 2, such a two-stage CTLE configuration faced challenges related to significant power consumption, primarily due to the intricate circuit topology. In the HFF-CTLE, the high-frequency feedforward module consumes a current I_H, and the main input transistor current consumption is $2I_M$. To ensure an adequate adjustment range for high-frequency peaking, the tail current source I_H should not be significantly smaller than $2I_M$. In the LFEQ module, as I_L is in the feedback path, a smaller current source is needed. The I_{BUF} influences the output common-mode voltage and swing and should be comparable in magnitude to $2I_M$. Then, the power consumption of the traditional implementation with two stages of CTLE for achieving triple peaking is: $(I_H+4I_M+I_L+I_{BUF})\times 1V$, which is substantial for low-power AFE designs.

A. Circuit Description

To address this concern, we suggest consolidating three peaks into a single-stage CTLE, aiming to streamline the

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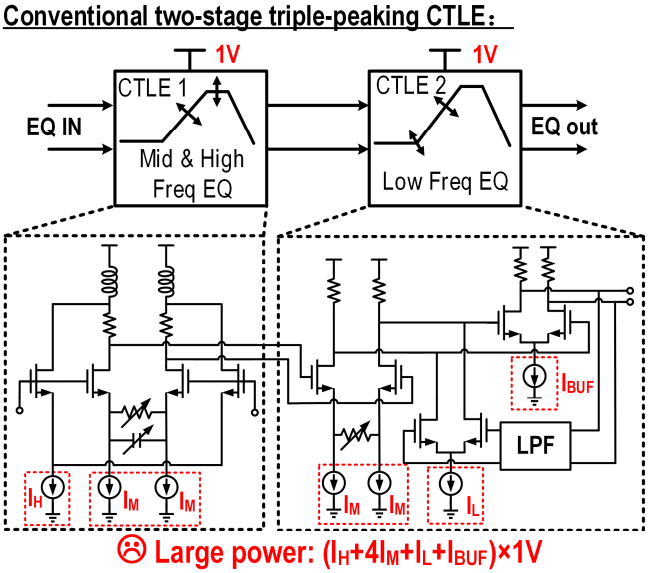


FIGURE 2. **Conventional two-stage triple-peaking CTLE.**

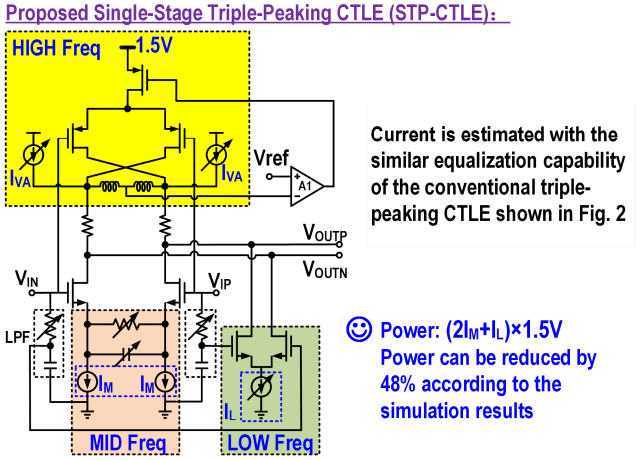


FIGURE 3. **Proposed STP-CTLE.**

circuit configuration, as demonstrated by our proposed STP-CTLE illustrated in Fig. 3. The high-frequency boosting component (HFB) of our STP-CTLE primarily incorporates a PMOS feedforward differential pair, effectively leveraging the entire current I_{TOT} ($I_{TOT}=I_M\times2+I_L$, see Fig. 3) of the STP-CTLE, thus leading to a significant power consumption reduction. High-frequency feedforward transistors act on the high-frequency gain through inductance without affecting the low-frequency gain[11]. As indicated in Fig. 3, since I_{TOT} also equals to (I_P+2I_{VAR}), the current of the PMOS differential pair I_P can be adjusted by tuning the current I_{VA} of two adjustable current sources. So, the transconductance of the PMOS differential pair in HFB part can be adjusted, thus the magnitude of the highfrequency peaking can be controlled. An operational amplifier A1 is used to form a common-mode feedback loop. Then, the point between the two inductors can be considered as a virtual ground. This allows the output of HFB to be directly coupled to the overall output of the STP-CTLE.

The mid-frequency peaking part (MFB) mainly consists of an RC source-degenerated differential pair. It adjusts the source-degenerated resistor and capacitor to tuning the position and magnitude of mid-frequency peaking[12]. In this setup, we employ a series resistor array and a parallel capacitor array to achieve the adjustment of resistance and capacitance values[13]. Additionally, increasing the source degeneration resistance in MFB results in a reduction of its transconductance (gm), which can enhance the linearity of the CTLE[14].

The foundation of the low-frequency boosting part (LFB) lies on a feedback loop, comprising a differential pair and a series of cascaded low-pass filters (LPF) integrated with a variable resistor R_{VAR}[15]. The tail current source of the differential pair is designed as an adjustable current source. Altering the size of the current source can modify the transconductance (gm) of the differential pair, thereby adjusting the strength of the low-frequency feedforward. By increasing the tail current I_L of LFB, the DC gain of the whole CTLE gets decreased while the high-frequency gain is not affected, thus reducing the low-frequency boosting. Moreover, the signal passes through a pair of RC low-pass filters before entering the gate of the differential pair. Consequently, low-frequency feedforward theoretically only acts on signals below the cutoff frequency of this low-pass filter (LPF). Since the resistance of the LPF is designed as an adjustable resistor array, altering the resistance values will change the cutoff frequency of the LPF, consequently influencing the location of low-frequency peaking.

Since the number of the stacked MOSFET is increased, the supply voltage for our STP-CTLE has been raised from 1 V to 1.5 V to ensure ample voltage headroom[13]. However, The power consumption of our proposed STP-CTLE is ($2I_M+I_L$) ×1.5V, which represents approximately a 48% reduction compared to the power consumption of the traditional multi-stage CTLE structure. This significant reduction is primarily attributed to the substantial decrease in current consumption within the STP-CTLE, which consequenced by the adoption of current-reuse techniques effectively saving the current overhead associated with the high-frequency feedback module. It's important to note that the power consumption comparison between the two is based on simulation results, assuming comparable equalization capabilities and bandwidth for both structures.

B. Zero and pole analysis

Fig. 4 depicts the small-signal equivalent circuit of the LFB and MFB for analyzing the locations of zeros and poles within its low-frequency and mid-frequency domain. The cutoff angular frequency of the Low-Pass Filter (LPF) in the figure is

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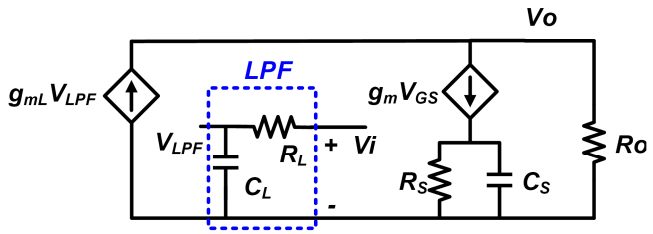


FIGURE 4. **The small-signal equivalent circuit of LFB & MFB.**

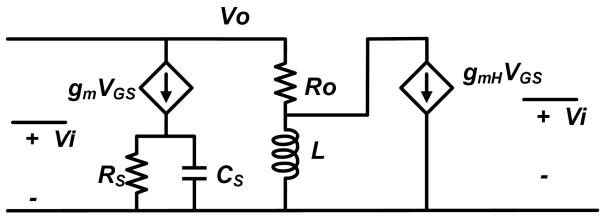


FIGURE 5.**The small-signal equivalent circuit of HFB & MFB.**

$$\omega_c = \frac{1}{R_L C_L} \tag{1}$$

The input V_{LPF} of LFB is obtained by passing V_i through the LPF. Therefore, we can straightforwardly deduce the pole generated by LFB to be

$$\omega_{p1} = \omega_c = \frac{1}{R_L C_L} \tag{2}$$

Then, by calculating the frequencies at which the sum of LFB and MFB outputs is zero, we can determine the zeros generated by LFB to be

$$\omega_{z1} = \frac{g_{mL}}{C_S(g_m - g_{mL})} - \frac{1}{R_S C_S} \tag{3}$$

The zeros and poles introduced by the source degeneration RC in MFB occur in the mid-frequency range, and they are given by:

$$\omega_{z2} = \frac{1}{R_S C_S},\ \omega_{p2} = \frac{1+g_m R_S}{R_S C_S} \tag{4}$$

Fig. 5 represents the small-signal equivalent circuit of HFB. Due to the introduction of inductance, the analysis of zeros and poles involves solving multiple equations. Assuming only HFB and MFB are presented (which does not significantly impact the zero-pole analysis since, despite the shared current, the outputs of HFB and LFB are superimposed on MFB, and there is no direct interaction between HFB and LFB), we obtain the transfer function for this scenario as follows:

$$H(s) = -\left[\frac{g_m(R_O+sL)}{1+g_m(R_S//\frac{1}{C_S})} + g_{mH}sL\right] \times \frac{1}{LC_Ls^2+R_OC_Ls+1} \tag{5}$$

It's worth noting that, as we analyze the HFB and MFB modules, there is a pair of zeros and poles here that coincide with the mid-frequency zeros and poles in LFB & MFB, denoted as ω_{z2} and ω_{p2}. Due to the inductance L and the load resistance R_O, forming an RLC model with the load capacitance C_L, two high-frequency poles will be generated. Additionally, gm_H introduces a high-frequency zero.

In equations (1)~(5), R_S and Cs represent the source degeneration resistance and capacitance of the equivalent half-circuit, respectively; gm, gm_L, and gm_H are the transconductance of the differential pairs of MFB, LFB and HFB, respectively; L is the inductance value, and C_L is the load capacitance of the CTLE.

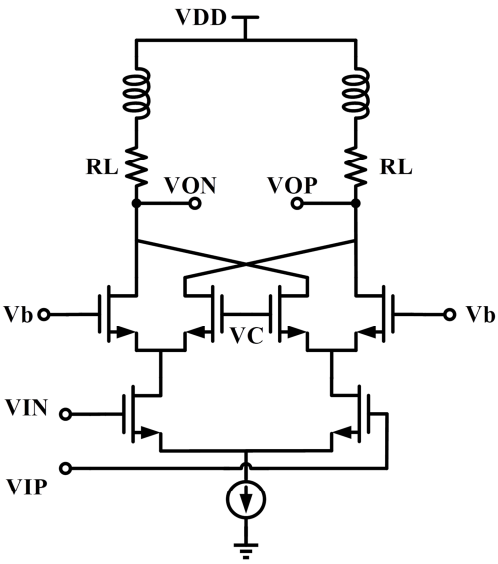


FIGURE 6. **The Circuit diagram of the Variable Gain Amplifier.**

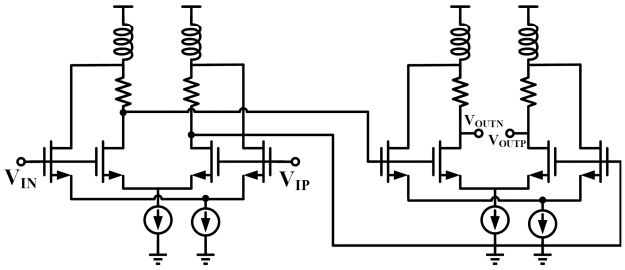


FIGURE 7. **The Circuit diagram of the Output Buffer.**

IV. Other Blocks

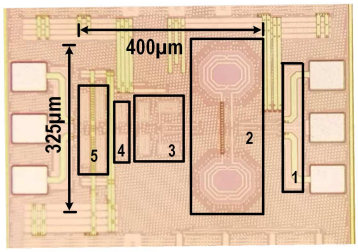
A. Variable Gain Amplifier

The Variable Gain Amplifier (VGA) adopted is an improved version based on the Gilbert cell as shown in Fig. 6. The traditional VGA structures include the Gilbert cell structure, where gain is controlled by adjusting the tail current source size, but it is challenging to achieve good linearity. Another approach is the source-degenerated differential pair, however, it suffers from significant bandwidth and group-delay variation across the gain settings. The VGA based on Transimpedance Amplifier (TIA) adjusts gain by changing the feedback resistor, but it has the drawbacks of structural complexity and high power consumption.

The VGA circuit in Figure 6 controls the conduction resistance of a pair of cross-coupled transistors by the voltage VC. Consequently, the magnitude of the alternating current flowing through the load resistor can be controlled.

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Power (mW)	
STP CTLE	12.633
BUF	4.0
VGA	4.482
Total	21.1m W

FIGURE 8. Chip photograph with power consumption breakdown.

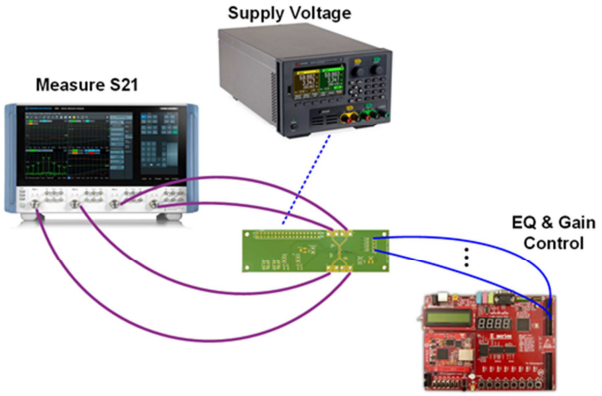


FIGURE 9. The platform for Chip testing.

When the voltage VC increases, the conduction resistance of the cross-coupled transistors decreases, resulting in a smaller alternating current through the load resistor RL, and hence, a smaller gain. The voltage VC is regulated externally using a Low Dropout Regulator (LDO).

B. Output Buffer

The output buffer is divided into two stages. The purpose of the first-stage pre-buffer is to ensure that the overall gain of the output buffer (OB) is not too small. This is necessary because the second-stage buffer serves to impedance match, having a smaller load resistor, which would result in a lower gain.

V. EXPERIMENTAL RESULTS

The proposed PAM4 AFE is fabricated in a 40-nm CMOS technology, occupying a core area of 0.136 mm2, as shown in Fig. 8. The measured power of the AFE except OB for testing is 21.1mW, corresponding to an energy efficiency of 0.33 pJ/bit. Fig. 9 illustrates the testing setup for the AFE chip, with S-parameters measured using a vector network analyzer. The internal parameters of the chip are controlled by an FPGA. The S21 parameter of the AFE, as shown in Figure 10, demonstrates that triple-peaking is achieved in a single stage CTLE with a maximum boost of 22.5-dB at 16-GHz Nyquist frequency.

As mentioned earlier, the magnitude of low-frequency peaking in the proposed AFE is controlled by the current source IL and the resistor RVAR in the LPF, while the

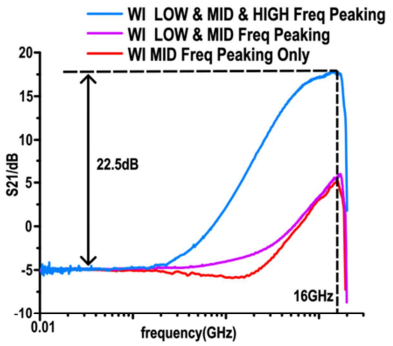


FIGURE 10. Measured S21 of the AFE.

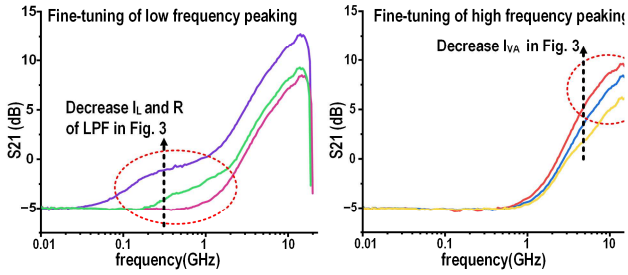


FIGURE 11. Measured fine tuning of low frequency peaking and high frequency peaking.

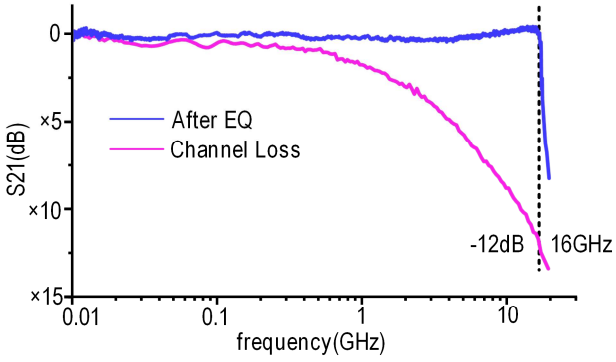


FIGURE 12. Measured AC response before and after channel loss compensation.

magnitude of high-frequency peaking is controlled by the current source IVA. The test results in Figure 11 illustrate the adjustment effects on the low-frequency and high-frequency peaking of the AFE. On the left side of Figure 11, the effects of adjusting IL and RVAR on low-frequency peaking are shown, while on the right side, the effects of adjusting the current source IVA on high-frequency peaking are depicted.

Next, we introduced a channel PCB between the AFE chip input and the vector network analyzer. By adjusting the parameters of the AFE, we obtained the S21 parameter of the AFE after channel equalization and compared it with the S21 of the channel PCB. As shown in Figure 12, the upper curve represents the S21 after equalization, and the lower curve represents the S21 of the channel. The

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TABLE I
PERFORMANCE COMPARISON

	This work	JSSC'18 [6]	VLSI' 18 [7]	TCAS-II'21 [14]	JSSC' 22 [16]	TCAS-I'22 [17]
Modulation	PAM4	PAM4	PAM4	PAM4	NRZ	PAM4
CMOS (nm)	40	28	16	40	28	65
Data rate(Gb/s)	64	64	56	56	56	56
Supply(V)	1.5/1	1	0.9/1.2/1.8	1.1/1.2	NA	1.1/1.2
Power (mW)	21.1	27$	34	30	43.6	28
Power Efficiency (pJ/bit)	0.33	0.42$	0.61	0.54	0.77	0.52
Equalization	22.5dB	16	17	9	16	14.37
Core Area (mm^2)	0.136	0.32	0.021	0.32	0.102	0.112

$CTLE Only

proposed AFE demonstrates effective channel equalization, achieving a compensation for a 12-dB channel loss at 16 GHz.

The measured performance of the RX AFE module is summarized in Table I and compared against recently published RX AFE for >56 Gb/s in CMOS technology nodes. At 64Gb/s data rate, the presented AFE display a minimum power efficiency of 0.33 pJ/bit. With the Equalization capability beyond 20dB, tht CTLE prove the best energy efficiency and triple-peaking capability.

VI. Conclusion

A PAM4 Receiver Analog Front-End operating up to 64 Gb/s has been presented. The AFE has been confirmed to possess the capability of triple peaking, with the maximum peaking reaching 22.5 dB. The power supply voltages are 1.5V and 1V, where 1.5V is the power supply voltage for STP-CTLE, and 1V is the power supply voltage for the other modules. Thanks to the proposed STP-CTLE, our AFE work achieves better energy efficiency and higher peaking boost at Nyquist frequency.

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